

dissatisfied with the features of Nagios software. Icinga developers released separate versions—Icinga Core, Icinga API and Icinga Web. In the following year, Icinga integrated the Icinga API component into Icinga Web and improvised SLA reporting. In 2012, the Icinga project released a technology preview of the core framework replacement and a parallel development branch called Icinga 2.

Icinga is an open source network monitoring tool that enables network administrators to monitor infrastructure of all sizes, with their integrated cluster system secured via SSL. It checks hosts and services and notifies users of their status. Icinga has a more agile development cycle than Nagios. It is an enterprise grade open source monitoring system which keeps a watch over networks as well as other network resources, notifies the user of errors and recoveries, and generates performance data for reporting. Scalable and extensible, Icinga can monitor complex, large environments over dispersed locations.

The following reasons ensure Icinga 2 remains one of the most popular open source network monitoring solutions:

- Good, stable and error-free code
- Interoperability with external interfaces
- High scalability
- Clean and clear—object based configuration
- Dynamic notifications
- All sorts of network logs
- Performs all sorts of service and agent-based checks

Features

Because of its regular agile based development, Icinga provides much better features than Nagios, apart from additional modules like improved SLA accuracy, and good database connections for PostgreSQL and Oracle in terms of network monitoring.

The following points highlight the unique features of the Icinga 2 network monitoring tool.

- **Extensive monitoring:** Icinga's network monitoring coverage is extensive in terms of network services, which include SMTP, POP3, HTTP, NNTP, Ping, etc. Icinga 2 also monitors host resources (CPU and disk utilisation, etc) and servers (all sensors, routers, switches, etc). It performs parallel service checks, and its simple front-end design helps administrators monitor the devices as per their unique requirements. Icinga 2 also supports Graphite and InfluxDB.
- **High performance and distributed:** Icinga 2 is very fast and can perform thousands of checks without any excessive CPU utilisation. It can combine high level clusters with distributed steps, and presents the complex network structure in a very simple way.
- **REST API:** With the support of RESTful API, Icinga 2 can update all sorts of configurations in real-time. Access control is highly configured and supports lots of functions, as per the number of users.

- **Interfaces:** Users can choose any of two interfaces like Classic UI and Icinga Web.
- **Innumerable modules:** Icinga 2 is bundled with many modules to provide advanced network monitoring and administration like Icinga Director, Business Process, Cube, Graphite, Elastic Search and Generic TTS, as well as other community modules like Grafana, Maps, PNP, etc.

Installing and deploying Icinga 2 and Icinga Web 2 on Ubuntu 17.10

Preparing your system

Before installing and configuring Icinga 2 and Icinga Web 2, you need the following:

- a. A machine running Ubuntu 17.10
- b. An IP address like 192.168.1.100
- c. All the installations of Icinga 2 and Icinga Web 2 done as the root user or as a user who is part of the sudo group

Part 1: Command based configuration of Icinga 2 and Icinga Web 2

Step 1: Update the repositories

In order to facilitate the installation of the latest version, it is very important to update the machine. Use the following command to update the repositories:

```
#apt-get update
```

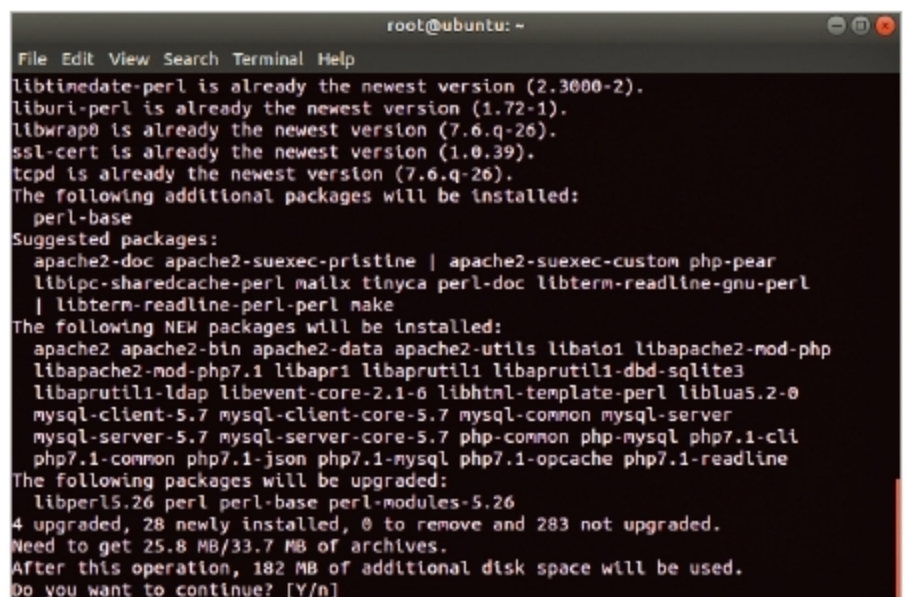
Step 2: Install the Net-tools package

To issue some network related commands on the console, like *ifconfig*-, in order to view the IP address, the Net-tools package is required. To install Net-tools, issue the following command:

```
#apt-get install net-tools
```

Step 3: Install the LAMP server

We need to have a machine with the LAMP server for the installation of Icinga 2. To create/install/deploy the LAMP



```
root@ubuntu: ~
File Edit View Search Terminal Help
libtimedate-perl is already the newest version (2.3000-2).
liburi-perl is already the newest version (1.72-1).
libwrap0 is already the newest version (7.6.q-26).
ssl-cert is already the newest version (1.0.39).
tcpd is already the newest version (7.6.q-26).
The following additional packages will be installed:
  perl-base
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom php-pear
  libipc-sharedcache-perl mailx tinycat perl-doc libterm-readline-gnu-perl
  | libterm-readline-perl-perl make
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libaio1 libapache2-mod-php
  libapache2-mod-php7.1 libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libevent-core-2.1-6 libhtnl-template-perl liblua5.2-0
  mysql-client-5.7 mysql-client-core-5.7 mysql-common mysql-server
  mysql-server-5.7 mysql-server-core-5.7 php-common php-mysql php7.1-cli
  php7.1-common php7.1-json php7.1-mysql php7.1-opcache php7.1-readline
The following packages will be upgraded:
  libperl5.26 perl perl-base perl-modules-5.26
4 upgraded, 28 newly installed, 0 to remove and 283 not upgraded.
Need to get 25.8 MB/33.7 MB of archives.
After this operation, 182 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Figure 1: Installation of the LAMP server on Ubuntu 17.10